

Analysis of UWA Communication using Bellhop simulator

Abstract

Simulation forms an important part of the development and empirical evaluation of underwater acoustic network (UAN) protocols. The key feature of a credible network simulation model is a realistic channel model. A common approach to simulating realistic underwater acoustic (UWA) channels is by using specialised beam tracing software such as BELLHOP. However, BELLHOP and similar modeling software typically require knowledge of ocean acoustics and a substantial programming effort from UAN protocol designers to integrate it into their research. This paper is a distilled tutorial on UWA channel modeling with a focus on network simulation, providing a trade-off between the flexibility of low level channel modeling via beam tracing and the convenience of automated channel modeling, e.g. via the World Ocean Simulation System (WOSS). The tutorial is accompanied by our MATLAB simulation code that interfaces with BELLHOP to produce channel data for UAN simulations. As part of the tutorial, we describe two methods of incorporating such channel data into network simulations, including a case study for each of them: 1) directly importing the data as a look-up table, 2) using the data to create a statistical channel model. The primary aim of this paper is to provide a useful learning resource and modeling tool for UAN protocol researchers. Initial insights into the UAN protocol design and performance provided by the statistical channel modeling approach presented in this paper demonstrate its potential as a powerful modeling tool for future UAN research.

Keywords: Underwater Communication, UWAC, BELLHOP.

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